AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

TM291
Management Information Systems

Midterm Examination (MTA)
Summer Semester 2025/2026

Time Allowed: 1.5 Hours

Instructions:

- Total Marks: **30**
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are not allowed.

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Answer ALL questions

Question 1: Select the Correct Answer (10 Marks)

1. Which of the following is considered a business objective for investing in information systems?
a) Increasing office space
b) Employee recreation
c) Improved decision making
d) Reducing product variety

2. What is a business model?
a) A method for storing business data
b) A type of database structure
c) A firm's hierarchy of management
d) How a company produces, delivers, and sells products and services

3. Which component forms the foundation upon which information systems are built?
a) Business processes
b) Social networking
c) IT Infrastructure
d) Collaboration tools

4. Which management level is primarily served by Executive Support Systems (ESS)?
a) Operational management
b) Middle management
c) Scientists and knowledge workers
d) Senior management

5. Which enterprise application helps manage relationships with suppliers and logistics companies?
a) CRM
b) ERP
c) KMS
d) SCM

6. The buying and selling of goods and services through the Internet is known as:
a) E-business
b) E-government
c) E-commerce
d) Social business

7. Which of the following is NOT one of the four basic business functions?
a) Human Resources
b) Finance and Accounting
c) Information Technology
d) Manufacturing and Production

8. In a relational database, which key uniquely identifies each record?
a) Foreign Key
b) Candidate Key
c) Attribute
d) Primary Key

9. Which relationship requires a join table?
a) One-to-one
b) One-to-many
c) Many-to-many
d) Recursive

10. What is the purpose of normalization?
a) Increase duplicate data
b) Store all data in one table
c) Minimize redundant data and awkward relationships
d) Eliminate primary keys

Question 2: Answer the following (10 Marks):

A. Differentiate between Data and Information (2 Marks).

B. What is a cross-functional business process? Provide one example (2 Marks).

C. Compare ERP and CRM systems in terms of their purpose and the business activities they support (3 Marks).

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D. Scenario: A company experiences delays in processing customer orders. The manager immediately buys a new software package without investigating the actual causes of the problem or considering alternative solutions.
Using the concepts of Critical Thinking, identify and explain three elements of critical thinking that the manager failed to apply.
(3 Marks)

Question 3: Design the following Database (10 Marks)

A university wants to create a database to manage students and the courses they register for. The database contains: STUDENT, Student_ID, Student_Name, Major, COURSE, Course_ID, Course_Name, Credits

A student can register for many courses and a course can have many students.

Required:

Design a simple ERD that resolves the relationship between STUDENT and COURSE (10 Marks).

You should:

- Identify the additional entity (join table for registration) (1 Mark).
- Show the primary and foreign keys (3 Marks).
- State the relationship type (3 Marks).
- Draw a clear ERD Representation (3 Marks

End of Questions

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